DO NOW — RECALL

Floating-point representation and normalisation

Name: _____ Class: _____ Date: _____

1. Write −2 in 8-bit two's complement.

2. What is unsigned 00100000 after a logical right shift by one?

3. Convert binary 0.1_2 to decimal.

4. Calculate 0.5×2^3.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

11111110.

2. Model answer

00010000.

3. Model answer

0.5.

4. Model answer

4.

Vocabulary — know these terms

mantissa — The signed significant part of a floating-point number.

exponent — The power of two scaling the mantissa.

normalised — A two's-complement fractional mantissa with different first two bits.

precision — Number of significant bits represented.

range — Spread of magnitudes that can be represented.

NEW LEARNING

Read, annotate and explain

Why is 0.0110 not normalised?

For a positive two's-complement fractional mantissa, a normalised value begins with 01, not 00.

Shift and compensate

Shift 0.0110 left one place to 0.1100. Decrease the exponent from 4 to 3 to compensate.

Verify the value

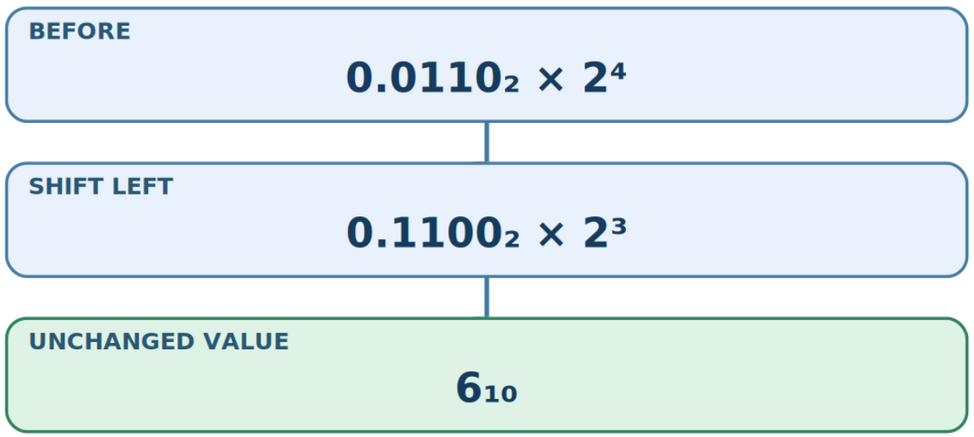
0.0110_2 is 0.375; 0.375 × 16 = 6. 0.1100_2 is 0.75; 0.75 × 8 = 6.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Read and interpret



BEFORE, $0.0110_2 \times 2^4$; SHIFT LEFT, $0.1100_2 \times 2^3$; UNCHANGED VALUE, 6_{10}

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Adjust and check

Exponent decreases from 4 to 3. Both representations equal decimal 5.

IMPROVE: Explain one correction to your own reasoning.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Shifting 0.0110×2^4 left to 0.1100×2^4 preserves its value.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

I DO — WORKED EXAMPLE

Follow the worked method

Normalise positive $0.0011_2 \times 2^5$.

First shift

0.0011 becomes 0.0110; reduce exponent 5 → 4. It still begins 00.

Second shift

0.0110 becomes 0.1100; reduce exponent 4 → 3. The first two bits are now 01.

Check

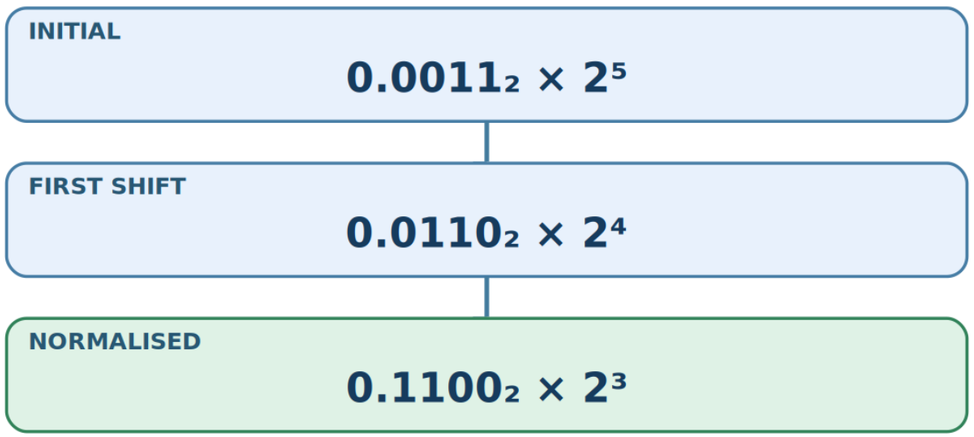
Both represent 6 in decimal, so normalisation preserved the value.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Normalise positive $0.0011_2 \times 2^5$.

0.0011 becomes 0.0110; reduce exponent 5 → 4. It still begins 00.



INITIAL, $0.0011_2 \times 2^5$; FIRST SHIFT, $0.0110_2 \times 2^4$; NORMALISED, $0.1100_2 \times 2^3$

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Why can normalisation improve precision without increasing the number of mantissa bits?

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Identify mantissa and exponent in $0.1100_2 \times 2^3$.

Question 2

Normalise positive $0.0010_2 \times 2^5$.

WE DO — GUIDED PRACTICE

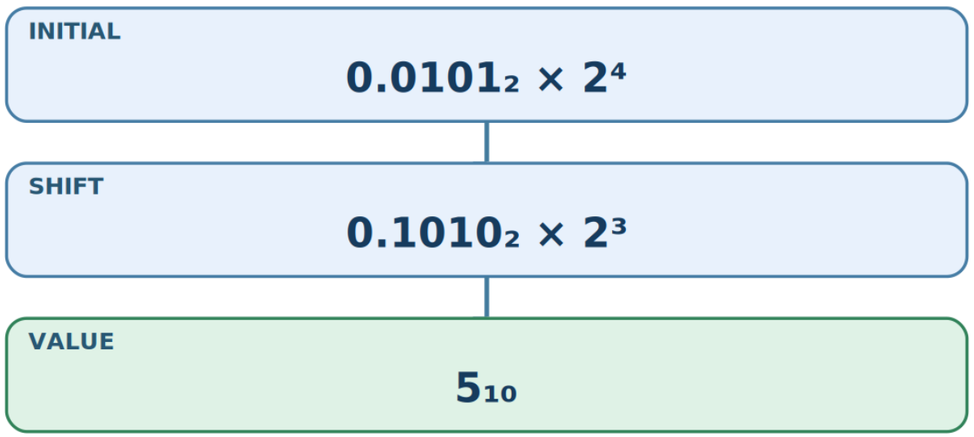
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Normalise positive $0.0101_2 \times 2^4$.

Shift the mantissa left one position: 0.0101 → 0.1010.



INITIAL, $0.0101_2 \times 2^4$; SHIFT, $0.1010_2 \times 2^3$; VALUE, 5_{10}

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict the shift Shift the mantissa left one position: 0.0101 → 0.1010.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.